TASK SPECIFICATION: P3-02 PHASE P3 — FRONTEND ARCHITECTURE (REACT 19 + TYPESCRIPT)

Internationalization (AR/EN), RTL/LTR Switching & Theme Toggle

Git Branch: `feature/P3-02-i18n-theme` · Target Pull Request: feature/P3-02-i18n-theme

Core Objective & Technical Scope:
Bilingual internationalization engine (Arabic/English) via react-i18next with bidirectional RTL/LTR layout adaptation and persisted theme toggle.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P3-02-i18n-theme`
Upstream Dependencies	P3-01		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Extract all interface strings into structured ar.json and en.json translation catalogs.
2. Configure I18nextProvider with dynamic document dir and lang attribute updates.
3. Implement ThemeToggle (dark/light) with local storage persistence.
4. Write tests verifying dynamic language and direction switching.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/shared/i18n/*`
- `locales/ar.json`
- `locales/en.json`
- `ThemeToggle/LangToggle`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Zero hardcoded strings in application shell.
- Document direction toggles between rtl and ltr seamlessly without broken layouts.

VERIFICATION & TEST COVERAGE STRATEGY

- I18n unit tests
- Playwright bilingual switching test